

# Sunghan Park

Engineering at Cambridge. Building tools for **crypto markets**, applied cryptography and ML.

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## EDUCATION

- University of Cambridge, Trinity College
- From Oct 2027
- Engineering, Information Engineering track
- Singapore American School
- Aug 2022 - May 2025
- High School Diploma. Mu Alpha Theta, Computer Science Honor Society, Science National Honor Society. Robotics, Research Club.

## WEB3 PROJECTS

- Arblineline · tennis and golf arbitrage scanner
- arblineline.vercel.app
- Compares tennis match and golf tournament prices across Polymarket, Kalshi and Pinnacle, including each venue's fees, and matches players across venues by name.
- Flags when backing both outcomes on two venues costs under \$1, and sizes the trade from the live order books.
- Vitals · memecoin health evaluator
- coin-vitals.vercel.app
- Grades Solana and EVM tokens 0–100 on mint/freeze authority, honeypot and tax checks, liquidity and LP locks, holder concentration (Gini, HHI, insider clusters) and trade flow, with hard caps for disqualifying risks.
- Curve Lab · secp256k1 cryptography from scratch
- secp-lab.vercel.app
- Curve arithmetic, Keccak-256, RFC 6979 ECDSA with ecrecover, BIP-340 Schnorr and Shamir sharing in JavaScript BigInt, verified against OpenSSL and published vectors. Live demo extracts a private key from a reused nonce.
- AMM Lab · Solidity constant-product pool and MEV simulator
- amm-lab.vercel.app
- Uniswap-V2-style pool in Solidity (OpenZeppelin, Hardhat). Simulator reads mainnet reserves over raw JSON-RPC and solves optimal sandwich and CEX-arbitrage trades; contract tests replay the sandwich and match the model to the wei.

## AI AND QUANT PROJECTS

- FinCouncil, open-source multi-agent LLM investment council: five analyst agents plus a CIO synthesis on live brokerage data, with cross-examination and fact-checking.
- QUANTDESK, paper-trading quant terminal (FastAPI): pluggable Python strategies, backtester, Binance and pump.fun memecoin feeds, alert scanner.
- Hold'em Noir, poker engine whose bots track Bayesian ranges over 1,326 combos, with Monte Carlo equity and a push/fold Nash solver by fictitious play.
- SwingLab, in-browser computer vision: MediaPipe pose and object detection, Hough-transform court detection, kinetic-chain analysis.

## EXPERIENCE

- Research Intern, UAV Research · A\*STAR Institute for Infocomm Research
- Apr 2024 - Aug 2024
- Ran Ubuntu and ROS 2 simulations to test and iterate drone perching mechanisms for forestry and aerospace tasks; co-developed a hybrid mounting leg and claw that improves quadcopter stability during aerial tasks.
- Co-Founder and Vice President · Sow4Future
- Dec 2023 - Present
- Youth financial-literacy and ESG-investing nonprofit reaching 2,000+ students in 15+ countries; lead a 30+ person team and launched competitions with 500+ participants a year and \$5,000+ in prizes.

## AWARDS

- Semifinalist, Wharton Global High School Investment Competition
- May 2024

## SKILLS

|            |                                                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------|
| CODE       | Python, JavaScript / TypeScript, Solidity, Java, SQL                                                                            |
| WEB3       | EVM and JSON-RPC, Hardhat, OpenZeppelin, AMMs and MEV, prediction-market microstructure, Solana token programs, ECDSA / Schnorr |
| ML / QUANT | MediaPipe, NumPy, pandas, Monte Carlo, Bayesian inference, backtesting, LLM agents, FastAPI, ROS 2                              |
| LANGUAGES  | English and Korean (native), Mandarin and French (professional)                                                                 |